Fostering websites accessibility: A case study on the use of the Large Language Models ChatGPT for automatic remediation

Achraf Othman
Mada Qatar Assistive Technology Center
aothman@mada.org.qa

Amira Dhouib
Mada Qatar Assistive Technology Center
adhouib@mada.org.qa

Aljazi Nasser Al Jabor
Mada Qatar Assistive Technology Center
aaljabor@mada.org.qa

ABSTRACT

The use of automated accessibility testing tools remains a common practice for evaluating web accessibility. However, the results obtained from these tools may not always provide a comprehensive and complete view of a site's accessibility status. The main purpose of this study is to improve web accessibility by automatically remediating non-accessible ones using Large Language Models (LLM), particularly ChatGPT. The effectiveness of the used model in detecting and remediating accessibility issues to ensure compliance with the Web Content Accessibility Guidelines (WCAG 2.1) is also discussed. By using ChatGPT as a remediation tool, this study investigates the potential of LLM in improving web accessibility. In the case study, two websites that did not adhere to the WCAG 2.1 guidelines were selected as the primary experimental subjects for the study. These websites were assessed using the web accessibility evaluation tool, WAVE, to detect accessibility issues. The identified issues served then as the basis for remediation using ChatGPT. The effectiveness of the used advanced language model as a web accessibility remediation tool was evaluated by comparing its findings with those obtained from manual accessibility testing. The results of this comparison have significant implications for stakeholders involved in achieving WCAG compliance and contribute to the development of more accessible online platforms for individuals with disabilities.

KEYWORDS

Web accessibility, Large Language Models, ChatGPT, Digital Accessibility

ACM Reference Format:
Achraf Othman, Amira Dhouib, and Aljazi Nasser Al Jabor. 2023. Fostering websites accessibility: A case study on the use of the Large Language Models ChatGPT for automatic remediation. In *Proceedings of the 16th International Conference on PErvasive Technologies Related to Assistive Environments (PETRA '23), July 05–07, 2023, Corfu, Greece.* ACM, New York, NY, USA, 7 pages. https://doi.org/10.1145/3594806.3596542

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
PETRA '23, July 05–07, 2023, Corfu, Greece
© 2023 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-0069-9/23/07…$15.00
https://doi.org/10.1145/3594806.3596542

1 INTRODUCTION

In recent years, web accessibility has become an increasingly prominent aspect in the field of digital design and development [1]. The aim is to ensure equal access to digital materials and services for People with Disabilities (PWD) [2]. Different automated tools exist to support the identification and remediation of web accessibility issues [3]. Despite their potential, the efficacy of these tools is somewhat constrained by their reliance on guidelines. Consequently, many websites still present significant accessibility barriers that prevent PWD from accessing and utilizing them [4]. Moreover, the readiness of accessible websites is necessary to ensure equal access to digital content [5]. However, manual accessibility evaluation and remediation can be time-consuming and resource-intensive and may not lead to fully accessible websites [6]. In this paper, we present a case study on the use of ChatGPT, a Large Language Model (LLM) [7], for the automatic remediation of web accessibility. We explored the effectiveness of this model as an automated tool for web accessibility remediation by comparing its findings with those obtained from manual accessibility testing. The case study involved the accessibility remediation of two websites with respect to the Web Content Accessibility Guidelines (WCAG 2.1). The accessibility issues of the selected websites were detected using the web accessibility evaluation tool, WAVE [8]. Our choice of using WAVE was influenced by different key factors. WAVE is widely recognized for its ability to provide an extensive assessment of websites' accessibility. Also, it addresses a multitude of accessibility issues and proposes suggestions for improvement. Once the areas of non-compliance were identified, the selected websites are then subjected to remediation using ChatGPT. The effectiveness and accuracy of the obtained outcomes are subsequently evaluated. The resulting findings could have important ramifications for web developers and designers attempting to comply with the WCAG guidelines to obtain more accessible websites. By adopting ChatGPT for web accessibility remediation, these stakeholders can potentially save time and resources, while increasing the accessibility of their websites.

The remainder of this paper is organized as follows. The second section presents a literature review of web accessibility and focuses also on the use of automated accessibility testing tools and ChatGPT for web accessibility remediation. The third section explores the potential of ChatGPT as an innovative tool for Web accessibility remediation. The following section presents the obtained results and provides a thorough analysis of these results. The fifth section is devoted to discussing and exploring challenges and limitations. The final section concludes and summarizes the main contributions of the study. It also highlights future research directions and potential advancements in the field of web accessibility.

2 LITERATURE REVIEW

2.1 Web Accessibility

Web accessibility is important for ensuring that PWD can access and use websites effectively [9]. This involves developing websites that are perceivable, operable, understandable, and robust, in accordance with the Web Content Accessibility Guidelines (WCAG) established by the World Wide Web Consortium (W3C) [3]. In many countries, web accessibility is a legal obligation, and non-compliance can result in substantial legal and financial penalties for businesses and organizations [10]. However, it is disconcerting to note that a considerable number of websites still have severe accessibility hurdles that prohibit PWD from using them [4]. The WCAG guidelines provide a set of technical requirements for web accessibility that attempts to make websites accessible [11]. These guidelines are structured into three degrees of conformance: A (lowest), AA (mid-range), and AAA (highest). Conformance at higher levels implies conformance at lower levels [12]. The standard addresses a range of accessibility challenges including, for instance, visual, auditory, motor, cognitive, and neurological impairments [13].

2.2 Web Accessibility Evaluation Tools and ChatGPT

Manual accessibility evaluation, crucial for ensuring inclusive web experiences, can be a resource-intensive process and may not guarantee entirely accessible websites. To address these challenges, developers have turned to automated accessibility testing tools that can efficiently identify and resolve accessibility issues. These automated tools are designed to scan websites for potential accessibility issues in line with WCAG guidelines, subsequently proposing detailed reports that highlight accessibility errors and possible solutions. By pinpointing these issues, developers can effectively address them, thereby ensuring a more universally accessible online experience for all users. Notable examples of automated accessibility testing tools include Axe, WAVE, and Google Lighthouse, all of which have gained considerable popularity in recent years [6].

ChatGPT, an advanced language model developed by OpenAI, has exhibited promising results across a wide array of Natural Language Processing (NLP) tasks. Its applications include language translation, rewriting, proofreading, question answering, and the development of conversational agents [7]. As ChatGPT was released relatively recently, its applications in the field of accessibility remediation remain relatively underexplored. To our knowledge, this is the first such attempt to use ChatGPT for remediating Web accessibility issues. Integrating ChatGPT into the accessibility remediation process offers an array of significant benefits. These encompass saving time and resources. Additionally, it promises a consistently high degree of accessibility, and enables more inclusive web environments that cater to the diverse needs and abilities of their users [14].

3 CHATGPT AS AN AUTOMATED TOOL FOR WEB ACCESSIBILITY REMEDIATION

As mentioned earlier, the primary goal of this study is to delve into the potential of ChatGPT as an automatic remediation tool for web accessibility issues. To achieve this objective, we have strategically chosen two websites that did not comply with the WCAG 2.1 guidelines. These websites serve as primary experimental subjects for the study. Both automatic and manual testing methods are adopted to identify accessibility issues on the two selected websites. Next, ChatGPT is trained in the WCAG 2.1 guidelines to address and resolve any accessibility issues present on the selected websites. The effectiveness of ChatGPT in achieving accessibility compliance is evaluated by comparing the discovered issues by ChatGPT with those found through manual testing.

The data collected from the accessibility testing tool is analyzed to identify accessibility issues on the selected websites. The results of the automated testing tool are compared with those of manual testing to evaluate their effectiveness in identifying accessibility issues. The data collected from the ChatGPT is evaluated then based on the number and severity of accessibility issues identified using WAVE tool. Figure 1 and Figure 2 showcase the Arabic version of the Qatar Airways website and the English version of the British Airways website, respectively. Both screenshots show a list of errors and warnings based on the guidelines of WCAG 2.1.

Table 1 shows the number of identified accessibility errors found on both websites along with additional information such as WCAG guidelines AA compliance. It also provides a comprehensive overview about the global rank and the number of users of the two websites as determined by SimilarWeb. The choice of websites, each in a different language, for the data sample is a crucial aspect of the study. This deliberate inclusion ensures a more comprehensive understanding of accessibility challenges and solutions across diverse language contexts.

The consideration of websites in multiple languages can contribute to a better understanding of the model's capabilities and limitations in addressing accessibility issues across multiple languages. Additionally, this consideration can assist in identifying the localization-related accessibility challenges, which can inform the development of tailored accessibility solutions that cater to the unique needs of PWD in various language communities. The data collected is then analyzed to identify the accessibility issues on the two selected websites. The data collected from ChatGPT is assessed based on the number and severity of accessibility issues identified and remediated by the WAVE tool.

4 RESULTS AND ANALYSIS

As mentioned earlier, the case study revolved around two prominent airline websites, namely: Qatar Airways, showcasing an Arabic version, and British Airways, operating in English. Both websites were found to have significant accessibility issues, such as the absence of alternate descriptions of images, insufficient color contrast, and challenges with keyboard navigation and form labeling. To evaluate the accessibility of both websites, we used the automated testing tool, WAVE. This tool revealed 25 accessibility issues on Qatar Airways website and 14 errors on British Airways website. However, it is important to note that automated testing tools have some limitations, as not all tools can detect every accessibility error. The findings of manual testing revealed several accessibility issues on both websites, including those that were missed by automated testing tools (e.g., the navigation path).

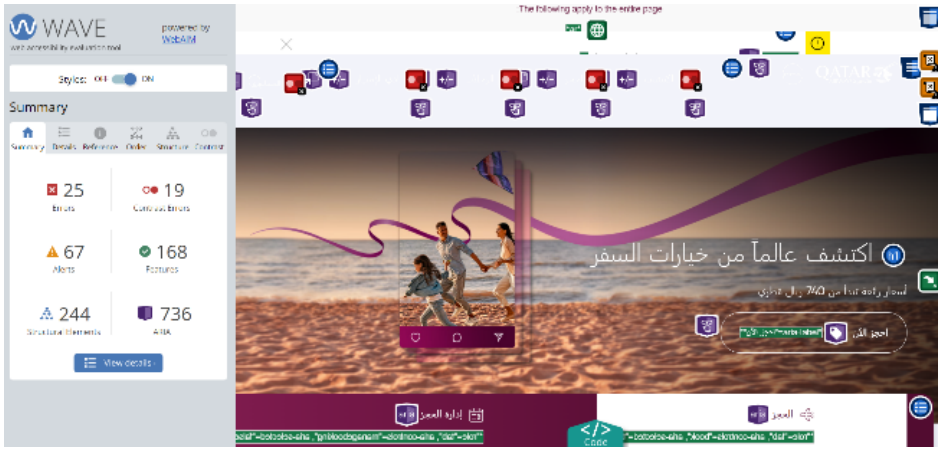


Figure 1: Web accessibility evaluation of Qatar Airways website (in Arabic language) using WAVE tool

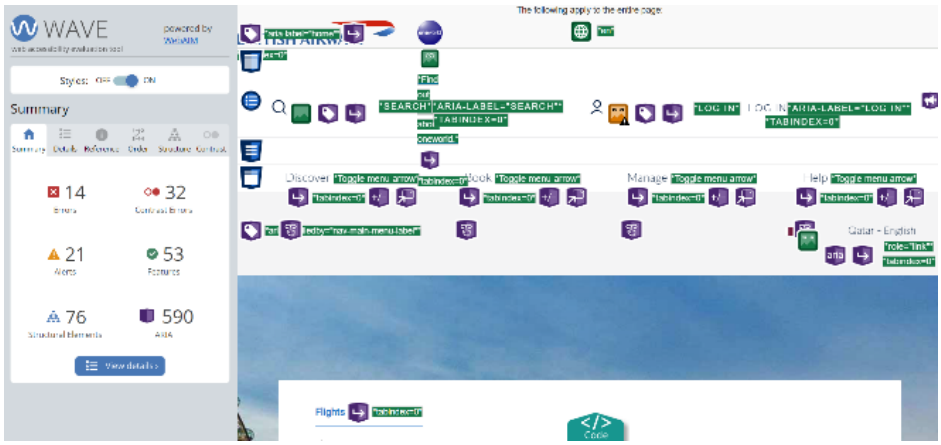


Figure 2: Web accessibility evaluation of the British Airways website (in English language) using WAVE tool

Table 1: Data sample overview

The selected Websites	WCAG guidelines AA compliance	Accessibility issues	SimilarWeb ranking
Qatar Airways website (Arabic version) [15]	No	25 errors and 59 alerts 28 contrast errors	Global Rank: 3617 Organic traffic: 13.5M (March)
British Airways website (English version) [16]	No	14 errors and 21 alerts 32 contrast errors	Global rank: 2582 Organic traffic: 15.9M (March)

By the use of ChatGPT, we were able to identify and remedy the accessibility errors on the selected websites. The inquiries were based on the WCAG 2.1 guidelines included in the HTML source code query. ChatGPT provided recommendations on how to resolve these issues and offered enhanced HTML source code to meet WCAG 2.1. The results of this study indicate that ChatGPT holds great promise as a tool for identifying and rectifying website accessibility issues. Notably, the used model was able to automatically resolve a considerable number of issues, reducing the manual effort required for remediation. However, it is important to acknowledge that there are several limitations to the use of ChatGPT in this context. For instance, the performance of the model depends on the quality of the training data and the accuracy of the WCAG guidelines. More details about the limitations will be discussed in the next section.

4.1 Accuracy

The study found that ChatGPT was highly accurate in remedying accessibility errors on both Qatar Airways and British Airways websites. Specifically, from a total of 39 errors that were initially

identified across these websites, ChatGPT successfully remediated 37 errors. This performance translates into a commendable accuracy rate of 94%. Based on these findings, it was demonstrated that ChatGPT proved to be an effective tool in resolving a considerable number of accessibility issues on the selected websites. Through its ability to generate HTML source code that met the WCAG 2.1 guidelines, it successfully addressed accessibility issues like non-descriptive link text, insufficient color contrast, and the absence of alternative image descriptions. Despite its substantial success in autonomously remediating accessibility issues, ChatGPT is not without its own set of limitations. For instance, subjective evaluations such as color contrast effectiveness may be challenging for the model to be determined accurately. Therefore, we recommend the adoption of a combination of automatic and manual testing methods to ensure that websites have the highest level of accessibility. In fact, combining different evaluation techniques and methods is necessary for most evaluation situations [17]. This necessity arises from the fact that there is no single technique able to fulfill all the evaluation constraints simultaneously.

4.2 Effectiveness

The effectiveness of the large language model ChatGPT in remediating accessibility issues was evaluated by comparing the number of issues that were automatically remediated to the total number of issues identified. The results showed that ChatGPT could remediate a significant number of issues automatically, reducing the manual effort required for remediation. However, not all issues were automatically remediated, and some issues required manual remediation. In addition, we highlight that if the given query is wrong, ChatGPT can identify that the success criteria are not linked to the tag or attribute. Below is an example of a query with a ChatGPT answer:

- Query: Given the following source code "«input title="Search input" id="search-modal-input" type="text" placeholder ="مثّل: نيويورك"/>", can you fix the accessibility issue related to the success criteria "Linked image missing alternative text" according to WCAG 2.1
- ChatGPT answer: There is no accessibility issue related to the success criteria "Linked image missing alternative text" in the provided source code, as there is no linked image present.

Table 2 provides more details about the errors found on the selected two websites, along with the results of remediation given by ChatGPT, and the manual testing suggested by an accessibility expert.

5 DISCUSSION AND LIMITATIONS

The findings of this study demonstrate the promising potential of ChatGPT as a tool for remediating web accessibility. Effectiveness analysis revealed that ChatGPT can automatically resolve a substantial number of issues, thereby reducing the amount of manual effort required for resolution. However, not all issues can be automatically addressed, and some require manual remediation. This study provides preliminary evidence of the potential of ChatGPT as a tool for web accessibility remediation. Additional research is required to determine the factors affecting the accuracy of the model and its applicability to different languages and website layouts.

Although the study yielded promising results, several limitations can be acknowledged. First, the study was limited to two websites, which may not be representative of the vast array of websites with different languages. To determine its generalizability, additional research is required to examine ChatGPT performance across a wider variety of websites. Additionally, certain accessibility issues could not be resolved, especially those that involve subjective assessment, such as the effectiveness of color contrast. Moreover, the research was limited to WCAG guidelines, and no other accessibility guidelines or standards were evaluated. To determine the applicability of ChatGPT to additional accessibility guidelines and standards further study is necessary, as websites, depending on their target audience and regulatory requirements, may need to comply with various supplementary guidelines and standards. Furthermore, the accuracy of ChatGPT in identifying and resolving accessibility issues may be affected by several variables, including the complexity of website's design. The model may be unable to detect certain types of issues and provide incorrect remediation suggestions. Hence, it is crucial to manually verify the obtained results manually and use ChatGPT in conjunction with other accessibility evaluation methods to ensure comprehensive and exhaustive outcomes. In addition, this study did not explore the impact of ChatGPT on website performance, such as loading times or page size. Large language models may generate additional code that could affect a website's performance, necessitating additional research on this matter.

6 CONCLUSION AND POSSIBLE SOLUTIONS

In this study, we examined the potential of ChatGPT as an effective and accurate tool for web accessibility remediation. The results showed that the used LLM is highly effective in resolving web accessibility issues, showcasing its potential to supplement the existing accessibility evaluation methods. In fact, we recommend using ChatGPT in conjunction with other accessibility evaluation methods, and manually checking the obtained outcomes to ensure accuracy. This study has practical implications for web accessibility professionals as it offers a promising solution for remediating website accessibility. ChatGPT has the potential to become a valuable tool for web accessibility testing and remediation, thereby increasing the accessibility of websites for individuals with disabilities.

This study lays the groundwork for future research on the use of ChatGPT as a large language model for web accessibility remediation. There are several directions for future research that could build on the findings of this study and expand the application of the LLM, such as ChatGPT, for web accessibility. First, future work could investigate the use of LLM on more complex websites, including websites with dynamic content, multimedia elements, and interactive features. Such websites may require more advanced NLP models and deeper integration with website codes to identify and remediate accessibility issues accurately. Furthermore, it could investigate the use of LLM in combination with other web accessibility evaluation methods such as manual testing and automated testing tools. The combination of different evaluation methods may provide a more comprehensive assessment of website accessibility and reduce the risk of false positives or negatives. Moreover, further

Table 2: Some errors found from WAVE tool and the corresponding remediation suggestions using ChatGPT and manual remediations

Error description	Manual remediation	HTML code tag	ChatGPT Query	ChatGPT remediation suggestions
Linked image missing alternative text	Add appropriate alternative text that presents the content of the image and/or the function of the link.		Given the following source code "$CODE$", can you fix the accessibility issue related to the success criteria "Linked image missing alternative text" according to WCAG 2.1	To fix this issue, we need to add an alternative text attribute to the img tag. This attribute should describe the content of the image in a way that can be understood by people who cannot see the image. Here's an example of how to add the alt attribute to the img tag:
Empty form label	If a text label for a form control is visible, use the <label> element to associate it with its respective form control. If there is no visible label, either provide an associated label, add a descriptive title attribute to the form control, or reference the label(s) using aria-labelledby. Labels are not required for image, submit, reset, button, or hidden form controls.	<input title="Search input" id="search-modal-input" type="text" placeholder= "مثل: نيويورك"/>	Given the following source code "$CODE$", can you fix the accessibility issue related to the success criteria " Empty form label" according to WCAG 2.1	The accessibility issue related to the success criteria "Empty form label" in the provided source code is that the input element does not have a visible label. To fix this issue, you can add a label element that is associated with the input element using the for attribute or wrap the input element with a label element that has a for attribute pointing to the id of the input element. Here's an example of how you can add a label element to the provided source code: <label for="search-modal-input">Search:</label><input title="Search input" id="search-modal-input" type="text" placeholder= "مثل: نيويورك"/>
Empty form label	Ensure that the form label contains text that describes the function of the associated form control. Labels are not required for image, submit, reset, button, or hidden form controls. If a label is not necessary visually, a descriptive title attribute may be added to the form control.	<label class="custom-control-label" for="economyclass"> </label>	Given the following source code "$CODE$", can you fix the accessibility issue related to the success criteria "Empty form label" according to WCAG 2.1	The accessibility issue related to the success criteria "Empty form label" in the provided source code is that the label element does not have any visible text or content. To fix this issue, you can add text or content within the label element that describes the purpose and function of the form element that it is associated with. For example, if the associated form element is a checkbox or radio button, you can add text that describes the option that the checkbox or radio button represents. Here's an example of how you can update the provided source code to include a visible label for a checkbox input element: <input type="checkbox" id="economyclass"><label class="custom-control-label" for="economyclass">Economy Class</label>

Multiple form labels	Ensure that at most one label element is associated to the form control. If multiple form labels are necessary, use aria-labelledby.	<input type="radio" id="mfirstclass" data-show= "الدرجة الأولى"value= "mfirst" data-value="mfirstclass" name="mcustomRadioCabin" class="custom-control-input"/>	Given the following source code "$CODE$", can you fix the accessibility issue related to the success criteria "Multiple form labels" according to WCAG 2.1	The accessibility issue related to the success criteria "Multiple form labels" in the provided source code is that the input element does not have an associated visible label. To fix this issue, you can add a label element that is associated with the input element using the for attribute or wrap the input element with a label element that has a for attribute pointing to the id of the input element. Here's an example of how you can add a label element to the provided source code: <label for="mfirstclass">First Class</label><input type="radio" id="mfirstclass" data-show="الدرجة الأولى" value="mfirst" data-value="mfirstclass" name="mcustomRadioCabin" class="custom-control-input"/>
Empty heading	Ensure that all headings contain informative content.	<h1 id="qrcode" title="qrcode" onclick="loadImage ('bookÖ02Dheader Ö02Dimage', 'https: //uqr.to/qwpa')"></h1>	Given the following source code "$CODE$", can you fix the accessibility issue related to the success criteria "Empty heading" according to WCAG 2.1	The accessibility issue related to the success criteria "Empty heading" in the provided source code is that the h1 element does not have any visible text or content. To fix this issue, you can add text or content within the h1 element that describes the purpose and function of the element. Here's an example of how you can update the provided source code to include a visible text for the h1 element: <h1 id="qrcode" title="qrcode" onclick="loadImage('book-header-image','https://uqr.to/qwpa')">QR Code Image</h1> In this example, the h1 element has the visible text "QR Code Image" added within the element, which describes the purpose and function of the h1 element. This will improve the accessibility of the element for users of assistive technology.
Broken ARIA reference	Ensure the element referenced in the aria-labelledby or aria-describedby attribute value is present within the page and presents a proper label or description.		Given the following source code "$CODE$", can you fix the accessibility issue related to the success criteria "Broken ARIA reference" according to WCAG 2.1	The accessibility issue related to the success criteria "Broken ARIA reference" in the provided source code is that the aria-labelledby attribute on the div element with id="join-pc-modal" is referencing an element with id="exampleModalLabel", but there is no element with that id on the page. To fix this issue, you need to either add an element with id="exampleModalLabel" to the page and provide appropriate text for it, or remove the aria-labelledby attribute from the div element. If you want to add an element with id="exampleModalLabel", you can add the following code before the div element with id="join-pc-modal": <h2 id="exampleModalLabel">Join Privilege Club</h2> Then, you can update the aria-labelledby attribute on the div element with id="join-pc-modal" to reference this h2 element: This will ensure that the aria-labelledby attribute on the div element references a valid id on the page. Alternatively, if you do not want to add an element with id="exampleModalLabel", you can remove the aria-labelledby attribute from the div element:

Empty link	Remove the empty link or provide text within the link that describes the functionality and/or target of that link.	 	Given the following source code "$CODE$", can you fix the accessibility issue related to the success criteria "Empty link" according to WCAG 2.1	Yes, the accessibility issue related to the success criteria "Empty link" can be fixed by providing a meaningful link text. A screen reader user relies on link text to determine where a link will lead them, and an empty link does not provide any context. In this case, we can provide a descriptive link text inside the anchor tag to describe the purpose of the link. For example, we can use "Download our Android App" as the link text. So the updated code with the fix will look like this: Download our Android App

work could explore the use of ChatGPT for accessibility testing and remediation in various languages and websites in different cultural contexts. This could provide insights into the cross-cultural applicability of the LLM and the potential challenges in adapting the model to different languages and cultural contexts. Additionally, future research could investigate the impact of LLM on website performance, including its impact on loading times, page size, and other performance metrics. This could provide insights into the trade-offs between website accessibility and website performance and inform decision-making in website development and management. Further studies are required also to determine the factors that affect the accuracy of the model and its applicability to different languages and website layouts.

REFERENCES

[1] I. Abu Doush, K. Sultan, M. A. Al-Betar, Z. Almeraj, Z. A. A. Alyasseri, and M. A. Awadallah, "Web accessibility automatic evaluation tools: to what extent can they be automated?," CCF Trans. Pervasive Comp. Interact., Mar. 2023, doi: 10.1007/s42486-023-00127-8.

[2] P. Brophy and J. Craven, "Web accessibility," Library trends, vol. 55, no. 4, pp. 950–972, 2007.

[3] Web Content Accessibility Guidelines (WCAG) 2.1. https://www.w3.org/TR/WCAG21/ (accessed Mar. 15, 2023).

[4] H. Y. Abuaddous, M. Z. Jali, and N. Basir, "Web accessibility challenges," International Journal of Advanced Computer Science and Applications (IJACSA), 2016.

[5] A. Othman, A. Al Mutawaa, A. Al Tamimi, and M. Al Mansouri, "Assessing the Readiness of Government and Semi-Government Institutions in Qatar for Inclusive and Sustainable ICT Accessibility: Introducing the MARSAD Tool," Sustainability, vol. 15, no. 4, p. 3853, 2023.

[6] A. N. Al Jabor, F. Adnan, M. Park, and A. Othman, "Mada Web Accessibility Monitor Tool," in 2021 8th International Conference on ICT & Accessibility (ICTA), Dec. 2021, pp. 1–5. doi: 10.1109/ICTA54582.2021.9809423.

[7] "Introducing ChatGPT." https://openai.com/blog/chatgpt (accessed Mar. 15, 2023).

[8] "WAVE Web Accessibility Evaluation Tools." https://wave.webaim.org/ (accessed Mar. 15, 2023).

[9] Y. Yesilada, G. Brajnik, M. Vigo, and S. Harper, "Understanding web accessibility and its drivers," in Proceedings of the cross-disciplinary conference on web accessibility, 2012, pp. 1–9.

[10] M. Paris, "Website accessibility: a survey of local e-government websites and legislation in Northern Ireland," Universal access in the information society, vol. 4, pp. 292–299, 2006.

[11] J. T. Richards and V. L. Hanson, "Web accessibility: a broader view," in Proceedings of the 13th international conference on World Wide Web, 2004, pp. 72–79.

[12] J. Kuzma, D. Yen, and K. Oestreicher, "Global e-government web accessibility: an empirical examination of EU, Asian and African sites," 2009.

[13] J. Lazar, A. Dudley-Sponaugle, and K.-D. Greenidge, "Improving web accessibility: a study of webmaster perceptions," Computers in human behavior, vol. 20, no. 2, pp. 269–288, 2004.

[14] D. Baidoo-Anu and L. Owusu Ansah, "Education in the Era of Generative Artificial Intelligence (AI): Understanding the Potential Benefits of ChatGPT in Promoting Teaching and Learning," Available at SSRN 4337484, 2023.

[15] "Qatar Airways." https://www.qatarairways.com/ (accessed Mar. 19, 2023).

[16] "British Airways." https://www.britishairways.com/ (accessed Mar. 19, 2023).

[17] A. Dhouib, C. Kolski, and M. Neji, "Towards a Web-based Multi-Criteria Decision Support System for the Layered Evaluation of Interactive Adaptive Systems," Universal Access in the Information Society, Nov. 2021, doi: 10.1007/s10209-021-00850-y.